

PHASE 3 – PROJECT DESIGN

Project Title

Streamlining IT Procurement: Automating Standard Laptop Orders

1. Introduction

The Project Design Phase defines the structure, workflow, modules, user interface, and data flow of the proposed IT procurement automation system.

2. System Workflow

The basic workflow of the system is:

Employee → Laptop Request → Manager Approval → IT Review → Procurement → Order Processing → Completed

If a request is rejected, the system updates the status as Rejected.

3. System Architecture

The system consists of the following major layers:

User Interface → Application/Business Logic → Database

- User Interface: Used by employees, managers, IT, procurement, and administrators.
- Application Layer: Handles requests, approvals, validation, and workflow.
- Database Layer: Stores users, laptop details, requests, approvals, and order information.

4. User Roles

Employee

- Login to the system.
- View standard laptops.
- Submit laptop requests.
- Track request status.

Manager

- View employee requests.
- Approve or reject requests.
- Provide approval information.

IT/Procurement Team

- Review approved requests.
- Verify laptop requirements.
- Process the laptop order.
- Update procurement status.

Administrator

- Manage users and system data.
- Monitor overall activities.
- Maintain system information.

5. Main System Modules

The system is divided into the following modules:

1. User Management
2. Laptop Catalog
3. Laptop Request
4. Approval Management
5. Procurement Management
6. Request Tracking
7. Notification/Status Update

6. Database Design

The system may contain the following main entities:

- User – Stores user details and roles.
- Laptop – Stores laptop model and specifications.
- Laptop Request – Stores employee requests.
- Approval – Stores manager/IT approval details.
- Procurement – Stores order and procurement information.

The entities are connected through appropriate IDs and relationships.

7. User Interface Design

The proposed screens include:

- Login Page
- Employee Dashboard

- Laptop Selection Page
- Laptop Request Form
- Manager Approval Page
- Procurement Dashboard
- Request Tracking Page
- Admin Dashboard

The interface will be designed to be simple and easy to use.

8. Request Status Design

The request status will be updated throughout the workflow:

Submitted → Pending Approval → Approved → IT Review → In Procurement → Ordered
→ Completed

Rejected requests will be marked as Rejected.

9. Input and Output Design

Inputs

- Employee details
- Laptop selection
- Required specifications
- Business justification
- Approval information

Outputs

- Request confirmation
- Approval status
- Procurement status
- Order information
- Final completion status

10. Expected Outcome

The Project Design Phase provides a clear structure for developing the system. It defines the system workflow, user roles, modules, database entities, and user interface required for automated laptop procurement.

11. Conclusion

The proposed design provides a structured and automated approach to standard laptop ordering. The design will be used as a reference during the Project Development Phase.